

NovaRetail+ — Behavioral Factors Associated with Annual Revenue

TripleTen Data Analytics Bootcamp · 2024 Customer-Level Correlational Analysis

Correlational analysis of 15,000 customers of a Latin American e-commerce platform to identify which behavioral factors are most strongly associated with the annual revenue each customer generates. Every relationship was measured with the coefficient appropriate to its variable type — Spearman, point-biserial and Cramér's V — and Pearson was benchmarked against Spearman across all key pairs to quantify, rather than assume, the distortion introduced by outliers.

Business Question

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Which customer behavior factors are most strongly associated with the annual revenue generated?

- Which behaviors track revenue most closely — purchase frequency, engagement, or advertising spend?
- Does premium membership explain customer value, or only appear to?
- Does region determine device type, which would justify channel-specific campaign design?

Methodology

- Load the 2024 customer table (15,000 records × 12 columns), confirm zero missing values, and rename all twelve columns from Spanish to English.
- Classify every column as numerical, binary or categorical — variable type is what determines which correlation coefficient is valid.
- Profile distributions: three key behavioral variables are right-skewed with long tails, which rules out Pearson as the sole basis for the analysis.
- Compute Spearman across the key numeric pairs, then benchmark every pair against Pearson and difference them to measure the impact of outliers.
- Apply point-biserial to each numeric–binary pair (premium membership, churn) and report coefficients alongside their p-values.
- Apply Cramér's V, derived from a χ^2 contingency table, to the categorical pair (device type × region).

```
cols = ['monthly_visits', 'monthly_purchases', 'targeted_ad_spend', 'annual_revenue']

pearson_mat = df[cols].corr(method='pearson')
spearman_mat = df[cols].corr(method='spearman')

for var1, var2 in combinations(cols, 2):
    r_p = pearson_mat.loc[var1, var2]
    r_s = spearman_mat.loc[var1, var2]
    data.append([var1, var2, r_p, r_s, round(r_p - r_s, 4)])

corr, p_value = pointbiserialr(df[numeric_col], df[binary_col])

chi2, p, dof, expected = stats.chi2_contingency(table)
cramers_v = np.sqrt(chi2 / (n * (min(table.shape) - 1)))
```

Python / pandas / scipy sample (coefficient selection and the Pearson-Spearman comparison): representative, not complete — the final notebook combines this with the profiling, cleaning and visualization steps.

Key Finding (Context → Finding → Implication)

Context

The Growth and Retention team needed to know where to concentrate effort ahead of the 2024 close, across 15,000 profiled customers — 13.9% on a premium subscription, 15.1% already churned.

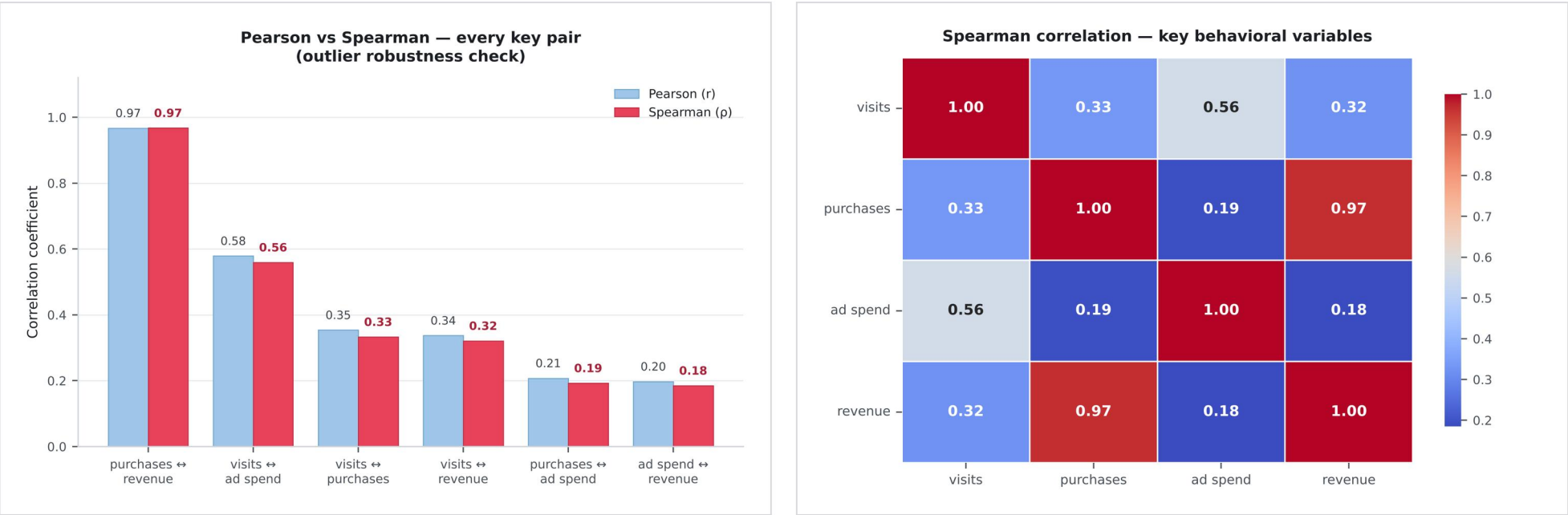
Finding

Purchase frequency is the dominant signal ($p = 0.97$). Advertising tracks visits ($p = 0.56$) far more than revenue ($p = 0.19$). Premium membership reaches $p \approx 3e-30$ with $r = 0.09$.

Implication

Optimize for conversion rather than traffic, judge campaigns on conversion-based return instead of spend, and drop premium status as a growth lever.

Visualization



Pearson vs Spearman for all six key pairs: outliers shift every coefficient, but only by 0.01–0.02.

Spearman correlation matrix for the four key behavioral variables.

Main Findings

Relationship	Coefficient	Value	Read
monthly_purchases ↔ annual_revenue	Spearman ρ	0.97	DOMINANT Strongest association in the dataset — and a metric-construction warning
monthly_visits ↔ targeted_ad_spend	Spearman ρ	0.56	SECONDARY Advertising is associated with traffic far more than with revenue
monthly_visits ↔ annual_revenue	Spearman ρ	0.32	SECONDARY Engagement relates to revenue, positively but loosely
targeted_ad_spend ↔ annual_revenue	Spearman ρ	0.19	WEAK Spend alone is a poor predictor of customer value
annual_revenue ↔ premium_member	Point-biserial r	0.09	NEGLIGIBLE $p \approx 3e-30$ and still too small to act on — $n = 15,000$
device_type ↔ region	Cramér's V	0.01	NO ASSOCIATION $p = 0.60$ — no basis for region-specific channel strategy

Significance vs relevance: premium membership reached $p \approx 3e-30$ against annual revenue and is still worth nothing as a lever: $r = 0.09$. With $n = 15,000$ any trivial effect clears a significance threshold, so effect size — not the p-value — is the decision-relevant quantity. **Collinearity flag:** the $p = 0.97$ between purchase frequency and annual revenue is strong enough to suggest the two metrics overlap in construction; it is reported as a validation task before predictive modeling, not as a finding. This is a correlational analysis — no result establishes a causal direction.